

Data Analyst Capstone Project Report

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IBM Data Analyst Professional Certificate



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OUTLINE



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EXECUTIVE SUMMARY



- Analyzed the Stack Overflow Developer Survey
- Compared current vs. future tech preferences
 - Programming languages
 - Databases & platforms
 - Respondent demographics
- JavaScript & PostgreSQL lead current use
- Cloud platforms dominate future plans
- Findings can guide hiring & training

INTRODUCTION



- Technology choices shift every year
- Teams need data, not guesswork, to plan
- This project explores what developers use today
- ...and what they plan to use next
 - Goal: spot emerging tools early
 - Audience: engineering leads & educators



METHODOLOGY



- Source: Stack Overflow Developer Survey
- Cleaned and grouped responses by category
- Built an interactive dashboard
- Compared "have used" vs. "want to use"
 - Languages, databases & platforms
 - Segmented by age, education & country



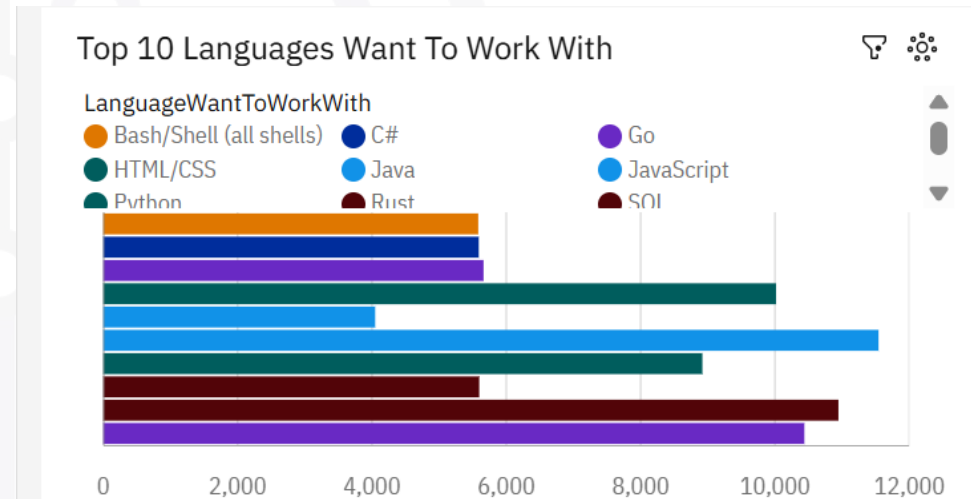
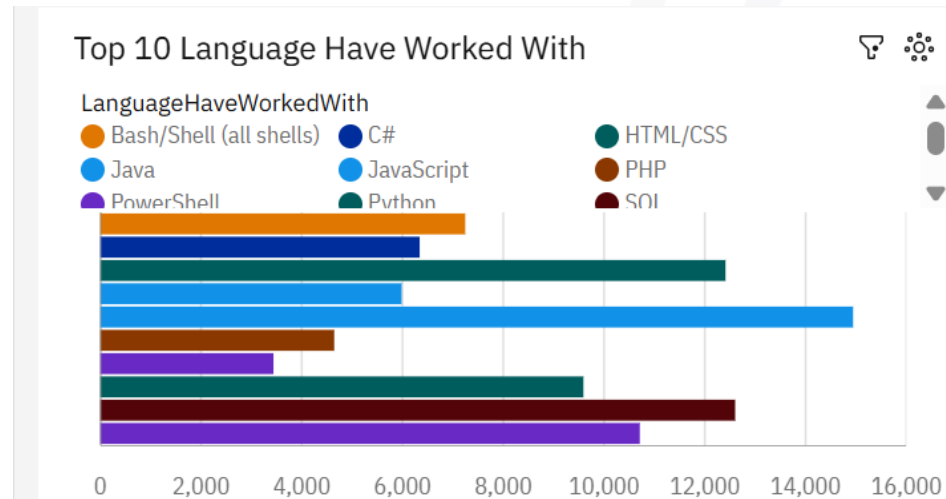
RESULTS



PROGRAMMING LANGUAGE TRENDS

Current
Year

Next
Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript remains the top language used
- Python & SQL hold steady in the top tier
- Go & Rust show the fastest growth in interest

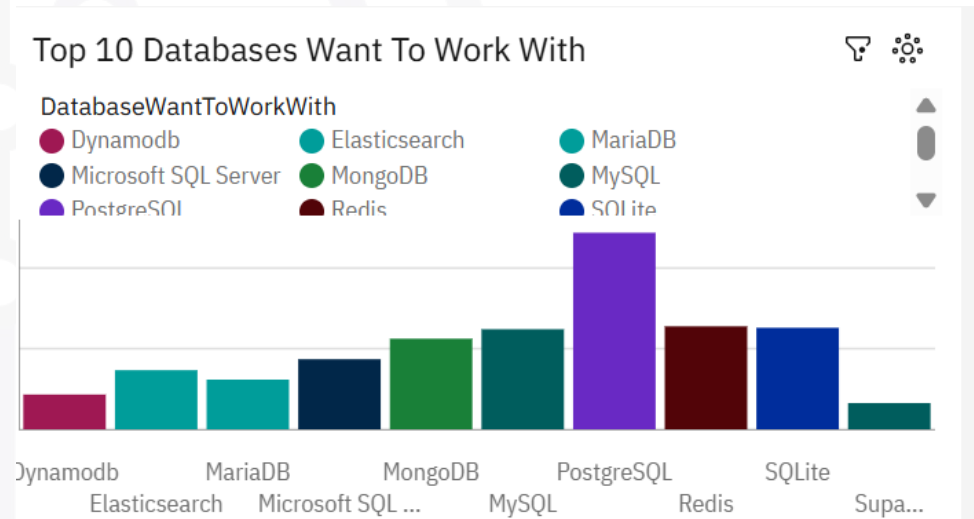
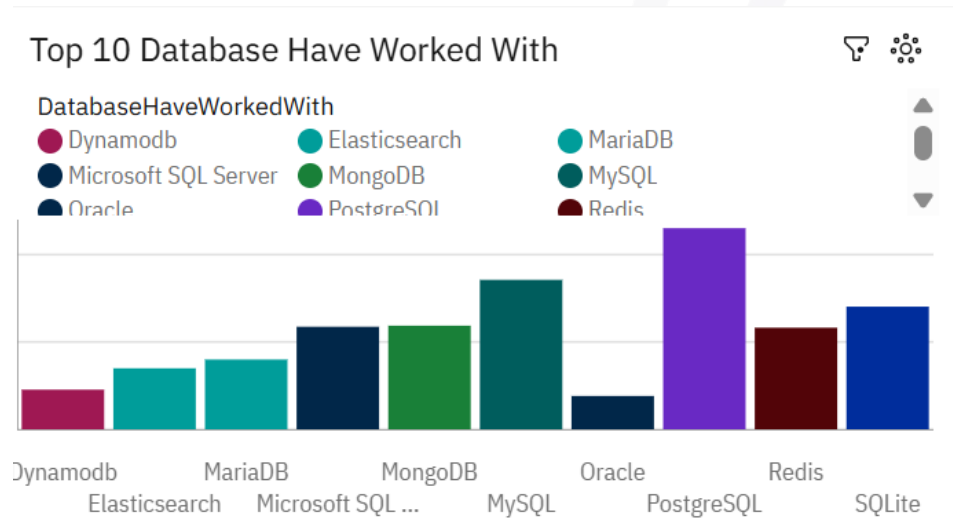
Implications

- Keep JS/Python training as core curriculum
- Start introducing Go & Rust in advanced tracks
- Monitor emerging languages each survey cycle

DATABASE TRENDS

Current
Year

Next
Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is the most-used database
- MySQL & MongoDB remain close behind
- PostgreSQL also leads future interest

Implications

- Prioritize PostgreSQL skills in hiring
- Keep MySQL/MongoDB support for legacy systems
- Low interest in legacy databases is a signal to migrate



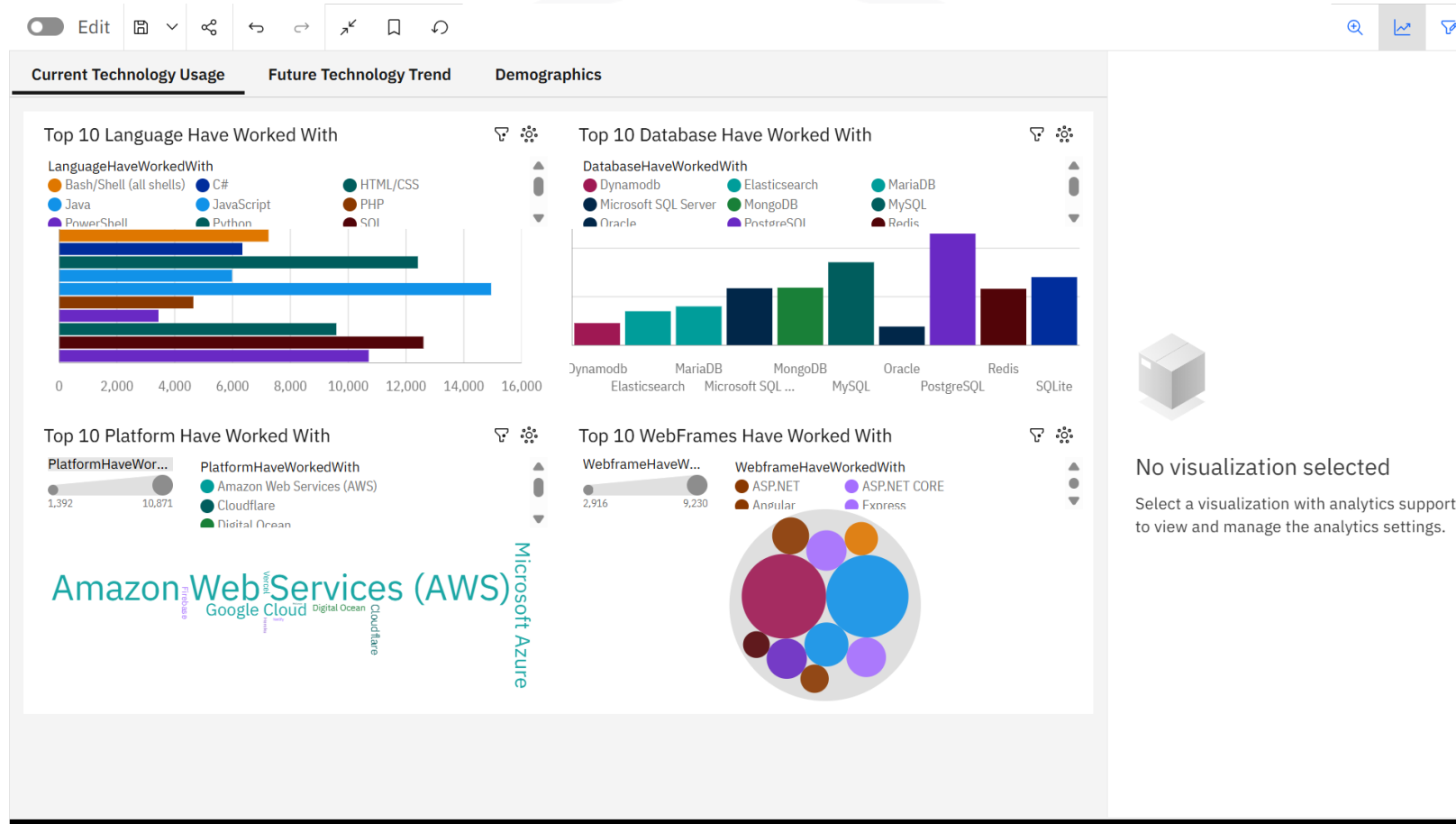
DASHBOARD



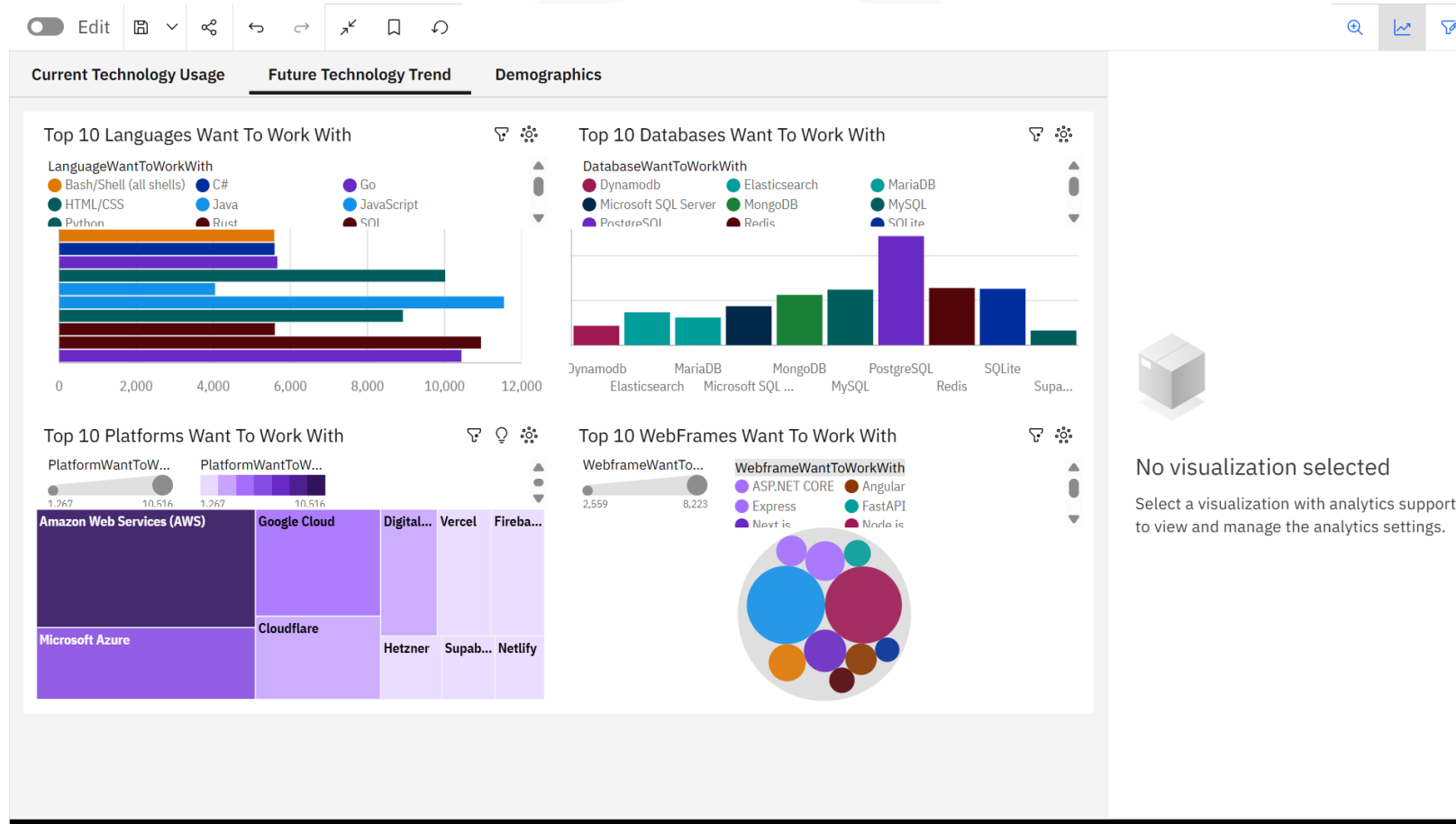
[View the live IBM Cognos dashboard →](#)



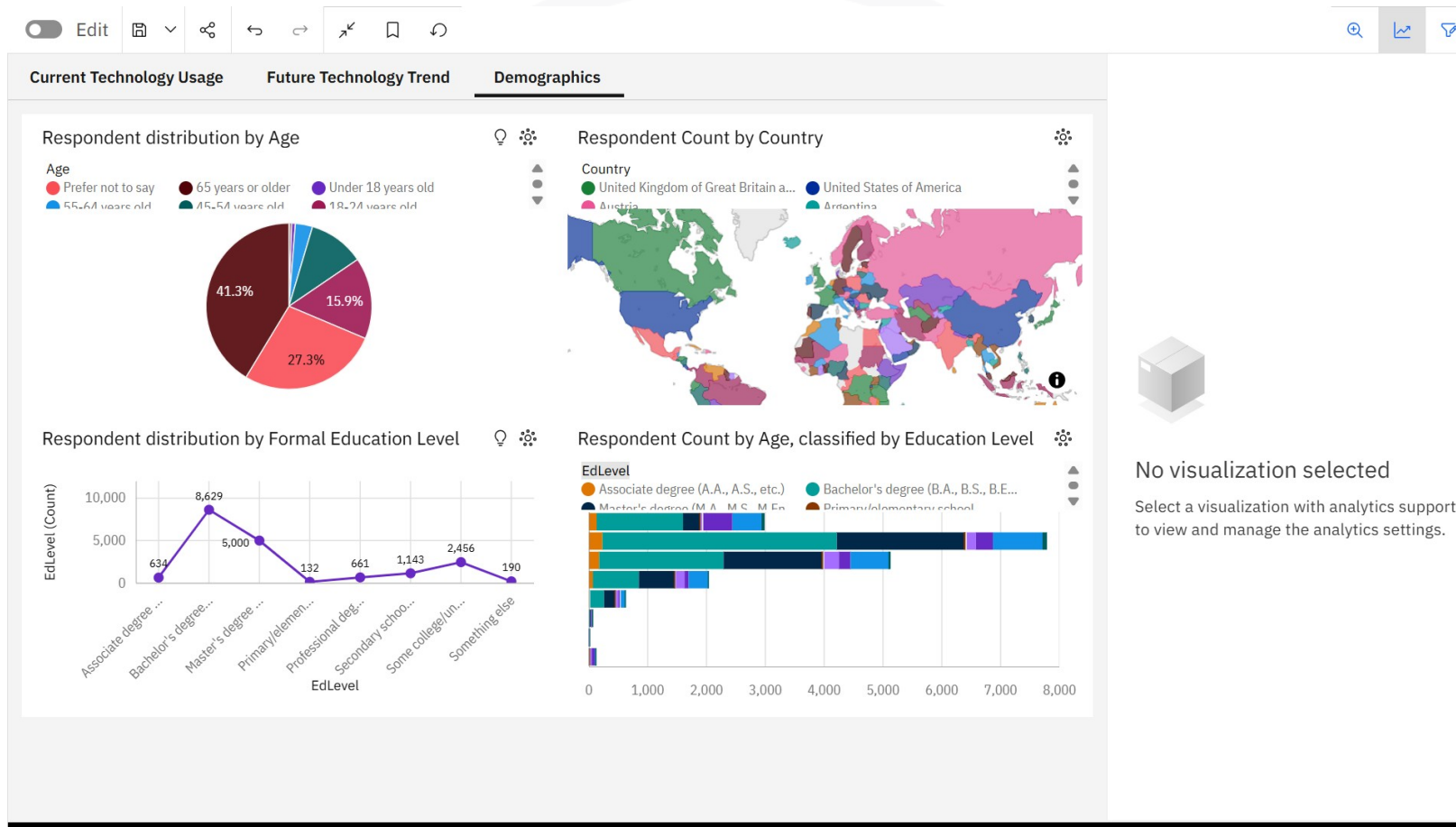
DASHBOARD TAB 1



DASHBOARD TAB 2



DASHBOARD TAB 3



DISCUSSION



OVERALL FINDINGS & IMPLICATIONS

Findings

- Core stack (JS, Python, PostgreSQL) stays dominant
- Cloud platforms are now the default choice
- Most respondents are early-to-mid career

Implications

- Anchor training around the core stack
- Invest further in cloud-native skills
- Design onboarding for a younger workforce



CONCLUSION



- The core developer stack remains stable
- Cloud & modern databases are rising fast
- Dashboard supports ongoing trend tracking
- Recommend revisiting this analysis yearly

APPENDIX



- No additional material — all key charts are covered in the Results and Dashboard sections.